Appendix A: Equilibrium and Linear-Stability Analysis of the Physics v1.0 Dynamical Sector

Supplementary Artifact v1.1 — Additive to Physics v1.0

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Abstract

Physics v1.0 introduces a three-variable phenomenological dynamical system and establishes its dimensional closure, but does not analyse it. This appendix supplies that analysis. We show that the system possesses a unique equilibrium in closed form; that the steady-state processing rate is fixed by input power alone and is independent of the entire entropy sector; that the equilibrium distance from the informational saturation parameter is set exactly by the irreversible-erasure rate; and that the system's only linear instability is a Hopf bifurcation whose threshold scales as the inverse of that same erasure rate, so that the system is unconditionally linearly stable in the zero-erasure limit. We also record two constraints absent from v1.0: an admissibility bound on the erasure rate, and a finite-time escape to unphysical states that requires an explicit domain restriction. All analytic results are verified numerically to machine precision.

This appendix is **additive**. It introduces no new postulate, modifies no equation, and has no bearing on the Gate 0 verdict.

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1 Scope and status

This document is a supplementary artifact. Under the immutability rules of the frozen research lineage:

1. Physics v1.0 remains v1.0. Nothing here edits it.
2. No equation, coefficient, or interpretation of v1.0 is altered. The system analysed below is exactly equations (9)–(11) of `paper_physics_v1.0_release.tex` as frozen.
3. No new postulate is introduced. Every statement is a consequence of the frozen ansatz.
4. Gate 0 Technical Report v1.0 and its verdict **UNRESOLVED IN GENERALITY** are untouched. Gate 0 audits the observer-entropy to horizon-entropy bridge; the present appendix analyses only the dynamical sector, which Gate 0 did not examine.

Remark 1 ($H_{\max}$ carries no geometric content here)**.** Throughout this appendix $H_{\max}$ is treated strictly as the *informational saturation parameter in bits* declared in v1.0's dynamical sector. Its identification with a causal-horizon capacity $A/(4\ell_P^2 \ln 2)$ is the structural domain extension whose microphysical justification v1.0 leaves open and Gate 0 finds unresolved. No result below depends on that identification, and none of them should be read as supporting it.

2 The system under analysis

Write $\dot{I} \equiv \dot{I}_{\text{model}}$ and $\dot{I}_e \equiv \dot{I}_{\text{erase}}$. The frozen system is

$$\frac{d\gamma}{dt} = a\,\dot{I}\,\mathcal{E}_{\text{avail}}\left(1 - \frac{\gamma}{\gamma_{\max}}\right) - b\gamma, \tag{1}$$

$$\frac{dH_{\text{obs}}}{dt} = c\gamma\,(H_{\max} - H_{\text{obs}}) - \dot{I}_e, \tag{2}$$

$$\frac{d\dot{I}}{dt} = e\,\frac{dH_{\text{obs}}}{dt} + f\mathcal{P}_{\text{input}} - g\dot{I}. \tag{3}$$

with the v1.0 dimensional assignments

$$[a] = [f] = \mathrm{J}^{-1}\mathrm{s}^{-1}, \qquad [b] = [e] = [g] = \mathrm{s}^{-1}, \qquad [c] = 1. \tag{4}$$

We treat $\mathcal{E}_{\text{avail}}$, $\mathcal{P}_{\text{input}}$ and $\dot{I}_e$ as constant control parameters, and all coefficients $a, b, c, e, f, g, \gamma_{\max}, H_{\max}$ as positive. It is convenient to introduce the *informational gap*

$$D \equiv H_{\max} - H_{\text{obs}}, \tag{5}$$

in terms of which (2) reads $\dot{D} = -c\gamma D + \dot{I}_e$.

3 Equilibrium structure

Proposition 2 (Unique equilibrium in closed form)**.** *For $\dot{I}_e > 0$ the system (1)–(3) admits exactly one equilibrium, given by*

$$\boxed{\dot{I}^* = \frac{f\mathcal{P}_{input}}{g}, \qquad \gamma^* = \frac{K\gamma_{\max}}{b\gamma_{\max} + K}, \qquad H^*_{obs} = H_{\max} - \frac{\dot{I}_e}{c\gamma^*},} \tag{6}$$

where $K \equiv a\,\mathcal{E}_{avail}\dot{I}^*$. Moreover $0 < \gamma^* < \gamma_{\max}$.

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Proof. At equilibrium $dH_{\rm obs}/dt = 0$, so the first term of (3) vanishes identically and (3) reduces to $f\mathcal{P}_{\rm input} - g\dot{I}^* = 0$, giving $\dot{I}^*$ as stated. Substituting into (1) with $d\gamma/dt = 0$,

$$K\left(1 - \frac{\gamma^*}{\gamma_{\max}}\right) = b\gamma^* \implies K = \gamma^*\left(b + \frac{K}{\gamma_{\max}}\right) \implies \gamma^* = \frac{K\gamma_{\max}}{b\gamma_{\max} + K}, \tag{7}$$

which is a single root, positive, and strictly below $\gamma_{\max}$ for finite K. Finally $dH_{\rm obs}/dt = 0$ gives $c\gamma^* D^* = \dot{I}_e$, hence $D^* = \dot{I}_e/(c\gamma^*)$ and $H^*_{\rm obs} = H_{\max} - D^*$. □

Corollary 3 (The entropy sector does not set the processing rate)**.** *$\dot{I}^*$ depends only on f, $\mathcal{P}_{input}$ and g. It is independent of $a, b, c, e, \gamma_{\max}, H_{\max}$ and $\dot{I}_e$.*

Remark 4 (Why no ignition threshold exists)**.** The three equations form a closed cycle: $\dot{I}$ drives γ through (1), γ drives $H_{\rm obs}$ through (2), and $H_{\rm obs}$ feeds back into $\dot{I}$ through (3). One might therefore expect a loop-gain condition separating a quiescent branch from a self-sustaining branch, i.e. bistability with an ignition threshold. No such structure is present. The reason is that the return path in (3) couples to $dH_{\rm obs}/dt$ rather than to $H_{\rm obs}$: a derivative coupling, which vanishes identically at any steady state. The loop is therefore open at equilibrium and closed only in the transient regime. This is a structural feature of the ansatz as written, not an artifact of parameter choice.

4 Admissibility constraints

Proposition 2 yields two constraints not stated in v1.0.

Proposition 5 (Erasure sets the equilibrium gap)**.** *At equilibrium,*

$$H_{\max} - H^*_{obs} = \frac{\dot{I}_e}{c\,\gamma^*}. \tag{8}$$

Consequently the v1.0 boundary condition $H_{obs} \le H_{\max}$ is satisfied with strict inequality whenever $\dot{I}_e > 0$, and is saturated exactly, $H^*_{obs} = H_{\max}$, in the limit $\dot{I}_e \to 0$.

Within the ansatz, irreversible erasure is therefore the sole mechanism holding the equilibrium state below the saturation parameter, and the equilibrium distance from saturation is inversely proportional to the relaxation rate γ^*. Under Remark 1 this is a statement about the model's internal bookkeeping and carries no horizon interpretation.

Proposition 6 (Admissibility bound on the erasure rate)**.** *Since H_{obs} is a von Neumann entropy in bits, $H_{obs} \ge 0$ is required. The equilibrium (6) is physically admissible only if*

$$\boxed{\dot{I}_e \;\le\; c\,\gamma^* H_{\max}} \tag{9}$$

Erasure rates above this bound drive the equilibrium value of H_{obs} negative and place the model outside its own domain of interpretation.

5 Linear stability

5.1 Jacobian and two identities

Linearising (1)–(3) about (6) in the coordinates $(\gamma, D, \dot{I})$ and writing

$$\alpha \equiv \frac{a\mathcal{E}_{\rm avail}\dot{I}^*}{\gamma_{\max}} + b, \qquad \beta \equiv a\mathcal{E}_{\rm avail}\left(1 - \frac{\gamma^*}{\gamma_{\max}}\right), \qquad \mu \equiv c\gamma^*, \qquad \delta \equiv cD^*, \tag{10}$$

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all strictly positive, the Jacobian is

$$J = \begin{pmatrix} -\alpha & 0 & \beta \\ -\delta & -\mu & 0 \\ e\delta & e\mu & -g \end{pmatrix}. \tag{11}$$

Two exact identities simplify everything that follows:

$$\boxed{\alpha\beta = a\,\mathcal{E}_{\text{avail}}\,b, \qquad \mu\delta = c\,\dot{I}_e} \tag{12}$$

The first follows from $\alpha = (K + b\gamma_{\max})/\gamma_{\max}$ and $\beta = a\mathcal{E}_{\text{avail}} b\gamma_{\max}/(b\gamma_{\max} + K)$; the second from $\delta = cD^* = \dot{I}_e/\gamma^*$ and $\mu = c\gamma^*$.

Note that α, β and μ depend on $\dot{I}_e$ not at all: by Proposition 2, neither $\dot{I}^*$ nor γ^* involves $\dot{I}_e$. The erasure rate enters the linearisation through δ alone, and does so linearly. This observation drives Proposition 8 below.

5.2 Characteristic polynomial

Expanding $\det(J - \lambda I)$ from (11), the off-diagonal structure causes the μ-dependent contributions of the third row to cancel against one another, leaving

$$(\alpha + \lambda)(\mu + \lambda)(g + \lambda) = \beta e \delta\, \lambda, \tag{13}$$

that is

$$\lambda^3 + \underbrace{(\alpha + \mu + g)}_{a_2}\lambda^2 + \underbrace{(\alpha\mu + \alpha g + \mu g - e\beta\delta)}_{a_1}\lambda + \underbrace{\alpha\mu g}_{a_0} = 0. \tag{14}$$

5.3 The Hopf threshold

Proposition 7 (Single stability boundary)**.** *The equilibrium (6) is linearly asymptotically stable if and only if $e < e_c$, where*

$$\boxed{e_c = \frac{(\alpha + \mu + g)(\alpha\mu + \alpha g + \mu g) - \alpha\mu g}{\beta\delta\,(\alpha + \mu + g)}} \tag{15}$$

At $e = e_c$ a complex-conjugate pair crosses the imaginary axis at frequency

$$\omega_c = \sqrt{\frac{\alpha\mu g}{\alpha + \mu + g}}. \tag{16}$$

Proof. For the cubic (14) the Routh–Hurwitz criterion requires $a_2 > 0$, $a_0 > 0$ and $a_2 a_1 > a_0$. The first two hold identically since $\alpha, \mu, g > 0$. The third, $(\alpha+\mu+g)(\alpha\mu+\alpha g+\mu g-e\beta\delta) > \alpha\mu g$, is linear and strictly decreasing in e because $\beta, \delta > 0$; solving for equality gives (15). At equality $a_1 = a_0/a_2$, so (14) factorises as $(\lambda + a_2)(\lambda^2 + a_0/a_2) = 0$, giving a real root $-a_2$ and a pair $\pm i\sqrt{a_0/a_2}$, which is (16). Since only a_1 depends on e and does so monotonically, the crossing is unique and transversal. □

Corollary 8 (Zero-erasure limit is unconditionally stable)**.** *Because α, β, μ are independent of $\dot{I}_e$ and $\delta = \dot{I}_e/\gamma^*$, the product $e_c\dot{I}_e$ is invariant under changes of $\dot{I}_e$:*

$$e_c \;\propto\; \frac{1}{\dot{I}_e}, \qquad \textit{equivalently} \qquad e_c = \frac{\alpha\mu\,[(\alpha + \mu + g)(\alpha\mu + \alpha g + \mu g) - \alpha\mu g]}{a\,\mathcal{E}_{avail}\,b\,c\,\dot{I}_e\,(\alpha + \mu + g)}, \tag{17}$$

using (12). Hence $e_c \to \infty$ as $\dot{I}_e \to 0$: in the zero-erasure limit the equilibrium is linearly stable for every finite feedback gain e, and the Hopf frequency (16) is unchanged.

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Remark 9 (Interpretation, stated narrowly)**.** Within this ansatz the only linear instability available to the system is driven by the irreversible-erasure channel. This is of interest because v1.0 explicitly declines to specify any relation between $\dot{I}_{\text{model}}$ and $\dot{I}_{\text{erase}}$ and records that omission as an open problem. The present result sharpens what is at stake in that omission: the unspecified relation controls not merely a dissipative cost but the existence of oscillatory dynamics. It does not supply the missing relation, and no relation should be inferred from it.

6 Numerical verification

All results were checked against direct numerical evaluation using the benchmark parameter set $a = b = c = f = g = \mathcal{E}_{\text{avail}} = 1$, $\gamma_{\max} = 10$, $H_{\max} = 100$, $\mathcal{P}_{\text{input}} = 5$, in the units of v1.0.

Quantity	$\dot{I}_e = 1$	$\dot{I}_e = 10$	$\dot{I}_e = 37$
γ^* (s^{-1}), Eq. (6)	3.3333	3.3333	3.3333
$\dot{I}^*$ (s^{-1}), Eq. (6)	5.0000	5.0000	5.0000
H^*_{obs} (bits), Eq. (6)	99.7000	97.0000	88.9000
Residual $\max\lvert\dot{\mathbf{x}}(\mathbf{x}^*)\rvert$	9.8×10^{-15}	4.4×10^{-16}	—
e_c analytic, Eq. (15)	44.880952	4.488095	1.212999
e_c numeric (eigenvalue crossing)	44.880952	4.488095	1.212999
$e_c\,\dot{I}_e$ (invariant, Cor. 8)	44.880952	44.880952	44.880952
ω_c analytic, Eq. (16)	0.925820	0.925820	0.925820
ω_c numeric	0.925820	0.925820	0.925820
$\max \operatorname{Re}\lambda$ at $e = e_c$	-1.5×10^{-16}	9.8×10^{-16}	-3.4×10^{-16}

Table 1: Analytic predictions against numerical evaluation. Agreement is to machine precision throughout.

7 Global behaviour and a required domain restriction

Linear stability is local. Direct integration of (1)–(3) shows that the basin of attraction is bounded and that trajectories leaving it do not merely diverge but reach *unphysical* states in finite time.

The mechanism is explicit in the equations. If γ is driven negative while the gap D remains positive, (2) makes dH_{obs}/dt strongly negative; (3) then drives $\dot{I}$ negative; and with $\dot{I} < 0$ and $(1 - \gamma/\gamma_{\max}) > 1$ the right-hand side of (1) is negative and growing in magnitude, so γ runs away. Nothing in the ansatz prevents this: the system has no positivity-preserving structure for γ or $\dot{I}$.

Numerically, with $\dot{I}_e = 10$ and an initial perturbation of 10^{-4} in γ:

Gain	Late-time behaviour
$e = 0.90\,e_c$	returns to (6); oscillation amplitude $< 10^{-7}$
$e = 1.01\,e_c$	bounded oscillation, γ amplitude $5.4 \times 10^{-3}\,\text{s}^{-1}$
$e = 1.02\,e_c$	bounded oscillation, γ amplitude $4.9 \times 10^{-2}\,\text{s}^{-1}$
$e = 1.04\,e_c$	escapes: $\gamma \to 0^-$ at $t \approx 259$
$e = 1.08\,e_c$	escapes: $\gamma \to 0^-$ at $t \approx 136$
$e = 1.20\,e_c$	escapes: $\gamma \to 0^-$ at $t \approx 55$

Table 2: Behaviour across the stability boundary. A bounded oscillatory regime exists in a narrow window above e_c; beyond it the trajectory leaves the physical domain in finite time.

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Accordingly the ansatz requires an explicit statement of its domain of validity. Two minimal options, either of which is a modelling choice rather than a derived result:

1. restrict the admissible parameter range to $e < e_c$ with Proposition 6 enforced; or
2. modify the ansatz so that $\gamma \geq 0$ and $\dot{I} \geq 0$ are invariant (for example by making the decay terms multiplicative).

Option 2 would change the frozen equations and therefore belongs to a future versioned artifact, not to this appendix.

8 What this appendix does and does not establish

Established. Within the frozen ansatz, and as consequences of it alone: uniqueness and closed form of the equilibrium (Prop. 2); independence of the steady-state processing rate from the entropy sector (Cor. 3); the erasure-set equilibrium gap (Prop. 5); the admissibility bound (Prop. 6); a unique Hopf boundary with closed-form threshold and frequency (Prop. 7); the exact inverse scaling of that threshold with the erasure rate (Cor. 8); and the finite-time escape requiring a domain restriction.

Not established. This appendix does not derive the ansatz from microscopic non-equilibrium dynamics; it analyses a system that v1.0 already labels phenomenological. It does not relate $\dot{I}_{\text{model}}$ to $\dot{I}_{\text{erase}}$. It does not give $H_{\max}$ any geometric or horizon meaning (Remark 1). It does not bear on the Gate 0 question, and its results must not be cited as progress on the observer–horizon mapping, which remains **UNRESOLVED IN GENERALITY**. It makes no cosmological prediction and does not constrain $(\epsilon, \beta, \phi_0)$ in v1.0's gravitational-wave template. It determines no numerical values for $a, \ldots, g$; the benchmark set above is illustrative only, chosen to exhibit the structure, and the qualitative conclusions are parameter-independent while the tabulated numbers are not.

Nearest open problems made concrete by this analysis. (i) Specify the $\dot{I}_{\text{model}}$–$\dot{I}_{\text{erase}}$ relation and recompute e_c, which by Cor. 8 then ceases to be a free parameter. (ii) Determine the criticality of the Hopf bifurcation analytically; the numerics above are consistent with a supercritical bifurcation whose limit cycle is rapidly destroyed, but no normal-form computation is offered here. (iii) Establish whether any positivity-preserving modification reproduces the same e_c scaling.

9 Reproducibility

All numerical results were produced by `physics_v1_0_ode_analysis.py`, accompanying this appendix, using NumPy 2.4.4 and SciPy 1.17.1. Equilibria are evaluated in closed form and verified by residual; stability boundaries are located both analytically via (15) and numerically by bracketing the sign change of $\max \operatorname{Re} \lambda(J)$; trajectories are integrated with LSODA at tolerances rtol $= 10^{-9}$, atol $= 10^{-12}$, with a terminal event on $\gamma = 0$. File hashes are recorded in the accompanying manifest.

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